

Age estimation based on speech features

Pasqualina Ambrosio
Politecnico di Torino
Student id: s345061
s345061@studenti.polito.it

Sonia Foco
Politecnico di Torino
Student id: s343043
s343043@studenti.polito.it

Abstract—This project focuses on building a regression pipeline to estimate the age of a speaker based on speech characteristics. In particular, the proposed methodology is based on the segmentation of the dataset into two subgroups according to the length of the samples and on the extraction of Mel and MFCC features from audio recordings. Various regression models are compared to identify the most effective ones and perform the evaluation.

I. PROBLEM OVERVIEW

The proposed task consists in building a regression pipeline capable of predicting the age of a speaker based on audio features. The evaluation metric for this task is Root Mean Square Error (RMSE), which measures the average deviation of predictions from the true values, with larger errors penalized more heavily.

The dataset is composed of 3,624 samples:

- a *development set* containing 2,933 labeled samples;
- an *evaluation set* containing 691 unlabeled samples.

Each sample corresponds to an audio recording of a spoken sentence, provided with both acoustic and linguistic meta-data, such as pitch, jitter, shimmer, and energy, as well as demographic information like gender and ethnicity. The target output is a numerical value representing the speaker's age.

Firstly, we can make some considerations about the dataset. Figure 1 shows the distribution of the target variable age in the development set. The problem is clearly unbalanced, since the majority of the speakers is concentrated in the age range of 15 to 25 years, while there are only a few speakers over the age of 60 and under the age of 10.

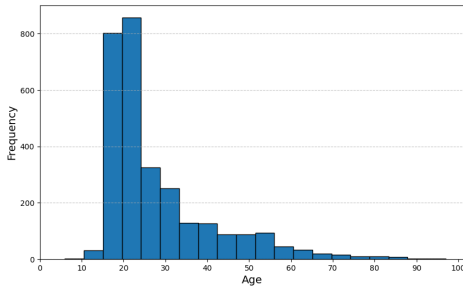


Fig. 1: Age distribution in the development set

Both the development and the evaluation set are homogeneously distributed in terms of gender, while they are heterogeneously distributed for ethnicities, since the majority of the speakers is Igbo or English.

All the audio recordings have been sampled with a sample rate of 22050 Hz, but they have different lengths. After listening to some of them and analyzing the distributions of the features *num words*, *num characters*, *num pauses* and *silence duration*, we observed that the dataset is composed of two main types of audios: short-duration samples with very brief and unclear spoken sentences, and longer-duration samples with more extended and clearer recurrent spoken sentences. Figure 2 shows the distribution of the number of words spoken in the whole dataset, showing that the longer-duration samples contain the same sentence composed by 69 words.

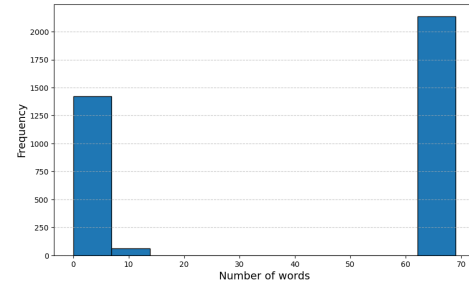


Fig. 2: Number of words spoken distribution in the dataset

We observed that most of the short samples are associated to Igbo speakers, while the longer ones are associated to different ethnicities like English, Spanish, Arabic, Mandarin, etc.

II. PROPOSED APPROACH

A. Preprocessing

The dataset does not contain any duplicate or missing value, therefore we did not have to handle such problems. We noticed an error in the *evaluation.csv* file, where the gender of one sample was labeled as "famale" instead of "female," and we corrected it. We then decided to remove the *sampling rate* feature, since it is the same for all the audios.

We proceeded to explore the relationships between the various features and age, aiming to identify those directly correlated with age and those that influence other parameters without directly impacting age itself.

To refine our approach, we opted to consider the gender feature only in its interaction with *spectral centroid mean*, as it appeared that females have a higher spectral centroid mean than males, accordingly with previous studies [1] and as we can see in Figure 3. To account for this, we encoded the

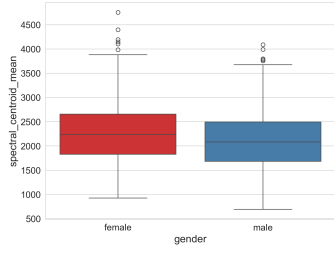


Fig. 3: BoxPlot of *spectral_centroid_mean* divided by gender

categorical *gender* feature with dummy variables composed of 0/1 values. Then, we created a new column representing the interaction of the male gender with the *spectral centroid mean* feature, effectively using the other category as the reference group. This resulted in the addition of the new feature *gender_male_spectral_centroid_mean_interaction*.

Since we can identify a well-defined distinction between the samples based on the number of words spoken, we decided to segment the dataset into two subgroups: short samples containing less than 69 words and long samples containing exactly 69 words. We will refer to these as *Short samples* and *Long samples* subgroups, respectively.

Additionally, we observed noticeable differences in the behavior of certain features between the two subgroups. One of the main differences is related to *ethnicity*, as the first subgroup is primarily composed of audio samples from Nigerian speakers, such as Igbos that represent the majority of the samples. Other significant differences can be found in the features *max_pitch*, *mean_pitch*, *hnr*, and *zcr_mean*, as these exhibit very different values between the two subgroups. Therefore, this segmentation seemed like a good choice, allowing us to train and evaluate models on more homogeneous subgroups, potentially improving performance and interpretability.

TABLE I: Cardinality of the subgroups

Subgroup	Development set	Evaluation set
Short samples	1223	263
Long samples	1710	428

In the *Long samples* subgroup, we further categorized speakers into two major groups according to their ethnicity: English and Others. As with the gender feature, we analyzed the interactions between these groups and other features correlated with age. In particular, we examined the interaction with *num pauses* and *silence duration*. As we can see in Figure 4 we noticed that these features are strongly related to the ethnicity of the speaker. Specifically, the English speakers exhibit lower silence durations and number of pauses than non-English speakers, likely due to their fluency in English, as the spoken sentences were in English. To address this, we encoded the new categorical feature *group* into 0/1 values and then added a new column representing the interaction between the Other group and the two features mentioned above. This resulted in the addition of the new features *group_other_silence_duration_interaction* and

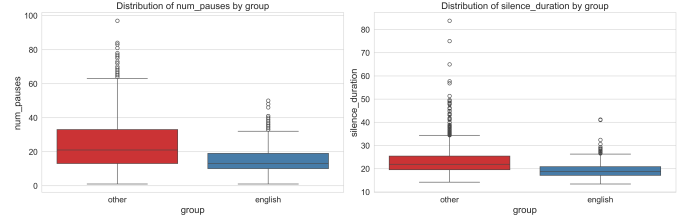


Fig. 4: BoxPlot of *num_pauses* and *silence_duration* divided by ethnicity categories

group_other_num_pauses_interaction for the *Long samples* subgroup, enabling us to remove the *ethnicity* feature. Since the uniformity in ethnicities in the *Short samples* subgroup, we did not have to manage such interactions, and we removed the *ethnicity* feature from the dataset.

Overall, this approach allowed us to capture nuanced relationships while maintaining a structured framework.

Before starting with the feature extraction process from the audios, we evaluated the duration of each audio file. The *Short samples* subgroup exhibited all short lengths. Most of those audios contained brief phrases, not always clear and in some case disturbed by background noise. For those reasons we decided not to extract audio features from those audios, considering the file with metadata already sufficient to perform our analysis. Whereas, the *Long samples* subgroup showed considerable variations. Therefore, shorter audio files were padded to reach a target length that we believed significant (35 seconds), while longer audio files were truncated.

Then, to extract potentially useful information from the audio files, we used the spectrograms of the signals and proceeded to extract Mel spectrogram and MFCC features, computing their mean and standard deviation for each audio. In general, the spectrogram of an audio is a visual representation of signal strength in terms of the signal’s loudness or intensity, which shows the variation of the frequencies over time. The Mel spectrogram is based on the representation of a signal spectrogram on the Mel scale, which is based on the human’s perception of certain sounds. It is a useful representation because it retains the information relevant to human perception. The overall process to extract the Mel spectrogram of an audio consists in dividing the signals into short overlapping windows, then converting them into the frequency domain using the Fourier transform and then the Mel filter banks are used to produce the spectrogram envelope [2]. In particular, we converted the Mel spectrogram to the logarithmic scale obtaining the Log-Mel spectrogram. On the other hand, the Mel Frequency Cepstral Coefficients are a set of coefficients that capture the spectral characteristics of an audio signal. They are derived from the Mel Spectrogram but are further processed to extract relevant information. [3] In particular, the process to extract MFCCs involves pre-emphasizing the input signal to remove unwanted or high frequencies, framing and windowing the signal, applying the

Fourier transform, using Mel filter banks to convert the signal to the Mel scale, computing the logarithmic magnitude of the power spectrum, and finally applying the discrete cosine transform (DCT). [2]

Figure 5, 6 and 7 respectively illustrate the representation of an audio signal in the time domain, the Log-Mel spectrogram and the MFCCs of the signal.

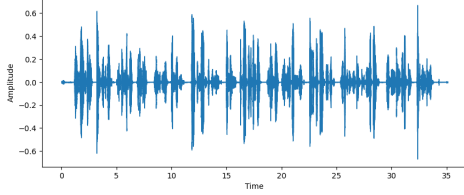


Fig. 5: Representation of an audio signal in the time domain

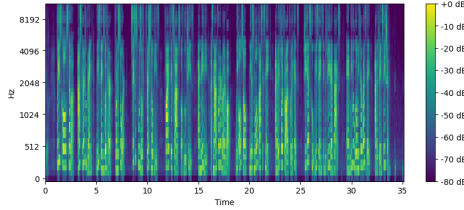


Fig. 6: Log-Mel spectrogram with $N_{mel} = 40$

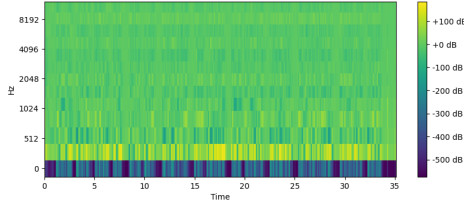


Fig. 7: MFCCs with $N_{mfcc} = 13$

We extracted the Log-Mel spectrogram and the MFCCs from each audio file in the *Long samples* subgroup using the librosa Python library. In particular, we used the parameter N_{mel} for the number of Mel frequency bands in the Mel spectrogram and N_{mfcc} for the number of MFCCs extracted from the audio. Then, we computed the mean and standard deviation of each one of the Mel frequency bands and MFCCs. This resulted in the extraction of $2 \cdot N_{mel}$ new features when using the Log-Mel spectrogram and $2 \cdot N_{mfcc}$ when using the MFCCs.

B. Model selection

We selected four models:

- *Linear Regressor*: it assumes a linear relationship between the input features and the target variable. It computes the coefficients that minimize the sum of squared residuals. It requires the standardization of the data

- *Ridge Regressor*: it is an extension of the *linear regression* that includes an L2 regularization factor that penalizes large coefficients more, trying to prevent overfitting. It requires the standardization of the data
- *Lasso Regressor*: it is an extension of the *linear regression* that includes an L1 regularization factor that tries to prevent overfitting and performs an implicit feature selection by driving to zero some coefficients. It requires the standardization of the data
- *Random Forest Regressor (RF)*: it combines the predictions from multiple decision trees, where each tree is trained on a random subset of the data and features.

We included the linear regression model as a baseline because it is one of the most fundamental models, making it a logical starting point for our analysis. To enhance model generalization, we also considered Ridge and Lasso regressors. Furthermore, to capture complex patterns and improve accuracy, we included the Random Forest Regressor: the ensemble step not only improves accuracy but also reduces the risk of overfitting [4].

C. Hyperparameters tuning

In our study, we aimed to enhance the performance of our model by identifying the optimal hyperparameters. The parameters to be tuned included:

- *Mel coefficients* and *MFCCs* in the feature extraction process of long-samples audios
- *Model-specific parameters*:
 - α for Ridge and Lasso regressors
 - $n_estimators$, $max_features$ for the Random Forest Regressor.

We selected hyperparameters (II) testing them on a 75/25 train/test split using a GridSearch in order to improve the model's predictive accuracy, using as scoring method the maximization of the negative root mean square error and performing a cross validation with $cv = 5$.

TABLE II: Hyperparameters

Model	Parameters	Values
Preprocessing	N_{mel}	20, 30, 40, 50, 60
	N_{mfcc}	10, 11, 12, 13, 14, 15, 16
Ridge	α	10, 15, 20
Lasso	α	0.05, 0.1, 0.5, 1
RF	$n_estimators$	{100, 200, 300, 400, 500, 600}
	$max_features$	{ "sqrt", "log2", None }

III. RESULTS

For the *Short Samples* dataset the best performance is obtained by Random Forest Regressor, obtaining satisfying results with an RMSE of 4.349.

When dealing with the *Long Samples* dataset, considering only the metadata of the csv file, we obtained an RMSE around 13 for all the models. So we decided to consider the features extracted from the audio files: in Figure 8 and 9 we can see the behaviour of the RMSE obtained by every model for different values of N_{mel} and N_{mfcc} . The lowest RMSE is

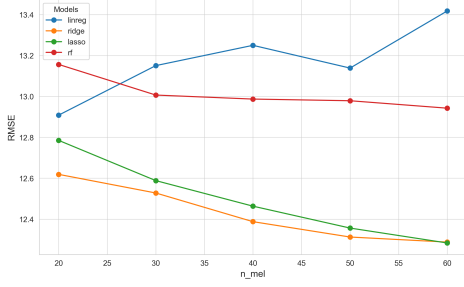


Fig. 8: Comparison of the performance of the considered models with different N_{mel}

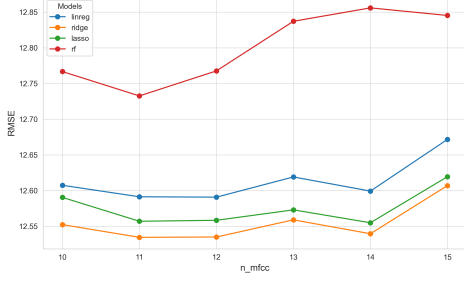


Fig. 9: Comparison of the performance of the considered models with different N_{mfcc}

obtained for the Mel representation using $N_{mel} = 60$ with Lasso Regressor, so the final model we selected is Lasso with $\{\alpha=0.1, \text{max_iter}=10000\}$ for the *Long samples* subgroup and RF with $\{\text{'max_features': 'sqrt', 'n_estimators': 600}\}$ for the *Short samples* subgroup. The performance of this model tested on a split of our complete development dataset divided in 80/20 train/test subsets is 9.527. In line with this result, the public leaderboard score is 9.585.

Since the performances of Lasso and Ridge in the *Long samples* subgroup are comparable, we decided to exclude the Ridge Regressor from further analysis. For the considerations we made in the Model Selection section, we instead decided to take into consideration the results of the Random Forest Regressor to reduce the risk of overfitting. Before applying this other model, we exploited the results obtained by Lasso and removed the zero importance features from our dataset. With this feature selection we improved the performance of RF Regressor with $\{\text{'max_features': None, 'n_estimators': 400}\}$, even though the performance of this model tested on a split of our complete development dataset divided in 80/20 train/test subsets is 10.204. Despite this, the improved evaluation obtained by the GridSearch with the cross validation led us to think that the model has actually improved, obtaining a public leaderboard score of 9.622. This can be due to the fact that the Random Forest Regressor is less affected by overfitting so the performance on the evaluation set is less likely to worsen.

IV. DISCUSSION

The proposed approach achieved satisfactory results, as it produced a lower RMSE compared to the public baseline.

In general, age prediction is a challenging task due to the unbalanced nature of the dataset.

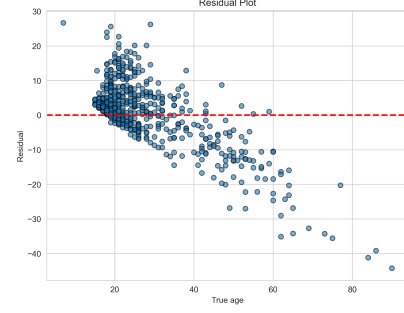


Fig. 10: Scatter Plot of the residuals vs the true ages of our test subset

As we can see in Figure 10, most speakers belong to the 15–25 age range, where the models seems to be quite well-trained, with a tendency to overestimate the ages for this group but provides a worse performance for the over-sixties where the predictions are considerably smaller than their real age. Consequently, this imbalance leads to higher RMSE scores. To overcome this problem, one possible improvement could be performing data augmentation of samples belonging to the minority age ranges. This consists in the generation of new audio samples by noise addition, time stretching, pitch shifting, or signal shifting of the original audios [5].

To further improve the results, more features could be extracted from the Log-Mel spectrogram and MFCCs. Indeed, each Mel frequency band and MFCC could be splitted into other n chunks resulting in a total of $n \cdot N_{mfcc}$ and $n \cdot N_{mfcc}$ chunks, from which statistical features can be computed.

Moreover, advanced deep learning models involving Deep Neural Networks can be implemented to achieve more accurate results. Studies have confirmed that deep learning approaches are effective in estimating the speaker's age [6].

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